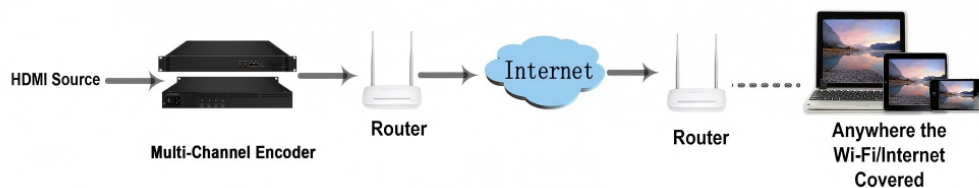


FMUSER FUQ-4335X-N Network Encoder 4-8-12 HDMI Inputs HDCP 1.4 HEVC-H.265 H.264 Encoding IP Output SRT-RTP-RTSP-RTMP-HTTP-HLS for CATV Satellite Terrestrial



Support IP out over UDP(SPTS) SRT/RTP/RTSP/RTMP/HTTP/HLS

Multiple HDMI inputs



Product Overview

The FMUSER FUQ-4335X-N is a multi-channel network encoder with configurable 4, 8, or 12 HDMI inputs supporting HDCP 1.4. It encodes video using HEVC/H.265 and H.264 formats at resolutions up to 1080P/720P 50/60Hz, including de-interlacing pretreatment. Output delivers IP streams via SRT, RTP, RTSP, RTMP, HTTP, and HLS over Ethernet, with web-based management. This device is ideal for CATV headends, satellite broadcasting, and terrestrial TV systems.

Key feature

Supports configurable 4/8/12 HDMI inputs with HDCP 1.4 compliance.

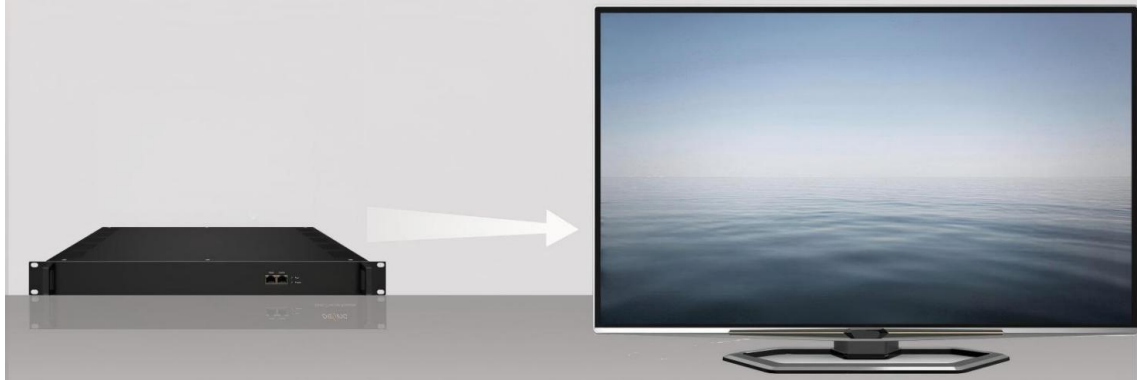
Features HEVC/H.265 and H.264 encoding with B-frame (IBBP) GOP for high compression.

Provides flexible IP streaming via SRT, RTP, RTSP, RTMP, HTTP, and HLS.

Ensures stable transport stream output with Fujitsu chip technology.

Offers advanced system management via web-based control.

OSD (Logo/QR Code) Insertion



Logo: support jpg/jpeg/bmp/gif/png file; drag to any position

QR Code: creat QR code quickly; add text/logo; custom size; drag to any position

- ◆ **Video Encoding:** HEVC/H.265 & MPEG 4 AVC/H.264
- ◆ **Audio Encoding:** MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough
- ◆ **Ultra Low Bit Rate:** Save 75% Bandwidth
- ◆ **Enhance Picture Quality:** Advanced Compressing Algorithm
- ◆ **Advanced Pretreatment:** De-interlacing, Noise Reduction, Sharpening
- ◆ **OSD (Logo, QR Code) insertion**—Optional as per order



News Channel/Movies

1Mbps Full HD



B frame(IBBP) GOP Structure



Sports Channel

2Mbps Full HD



HDMI 1.4



Full HD 1080P



HDCP 1.4



Decoder/STB

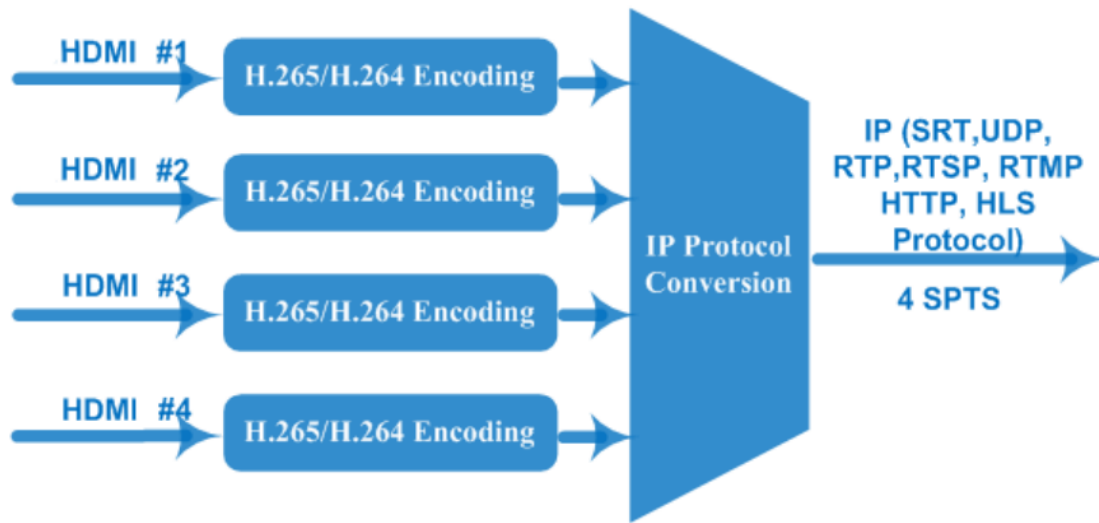
STB Available with Ensurity CAS

Decoding Chipset: Montage CS8051

NationalChip GX3201H

Principle Chart of Per Module

(The number of encoding channels depends on encoding format and resolution)



Technical Specification

Input	4/8/12 HDMI inputs for option, HDCP 1.4	4/8/12 HDMI inputs for option, HDCP 1.4	4/8/12 HDMI inputs for option, HDCP 1.4	4/8/12 HDMI inputs for option, HDCP 1.4	4/8/12 HDMI inputs for option, HDCP 1.4
Video Encoding	Encoding Format	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264	HEVC/ H.265, MPEG 4 AVC/H.264
Video Encoding	Resolution	Input	Input	Output	Output
Video Encoding	Resolution	Input	Input	HEVC/H.265	MPEG-4 AVC/H.264
Video Encoding	Resolution	4*1080P-50	4*1080P-50	4*1080P-50	2*1080P-50
Video Encoding	Resolution	4*1080P-60	4*1080P-60	4*1080P-60,	2*1080P-60
Video Encoding	Resolution	4*1080I-50	4*1080I-50	4*1080P-50	4*1080I-50 2*1080P-50
Video Encoding	Resolution	4*1080I-60	4*1080I-60	4*1080P-60	4*1080I-60 2*1080P-60
Video Encoding	Resolution	4*720P-50	4*720P-50	4*720P-50	4*720P-50

Video Encoding	Resolution	4*720P-60	4*720P-60	4*720P-60	4*720P-60
Video Encoding	Chroma	4:2:0	4:2:0	4:2:0	4:2:0
Video Encoding	Bitrate	0.5Mbps~20 Mbps(each channel)	0.5Mbps~20 Mbps(each channel)	0.5Mbps~20 Mbps(each channel)	0.5Mbps~20 Mbps(each channel)
Video Encoding	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video Encoding	GOP Structure	IBBP, IPPP	IBBP, IPPP	IBBP, IPPP	IBBP, IPPP
Video Encoding	Advanced Pretreatment	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening	De-interlacing , Noise Reduction, Sharpening
Audio Encoding	Encoding Format	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Pass-through	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Pass-through	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Pass-through	MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Pass-through
Audio Encoding	Sampling rate	48KHz	48KHz	48KHz	48KHz
Audio Encoding	Bit-rate (each channel)	48Kbps~384Kbps (MPEG-1 Layer 2& LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2& LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2& LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)	48Kbps~384Kbps (MPEG-1 Layer 2& LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)
Audio Encoding	Audio Gain	0~255	0~255	0~255	0~255

Stream output	IP (4 SPTS) output over SRT, RTP, RTSP, RTMP, HTTP, HLS per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (4 SPTS) output over SRT, RTP, RTSP, RTMP, HTTP, HLS per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (4 SPTS) output over SRT, RTP, RTSP, RTMP, HTTP, HLS per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (4 SPTS) output over SRT, RTP, RTSP, RTMP, HTTP, HLS per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter	IP (4 SPTS) output over SRT, RTP, RTSP, RTMP, HTTP, HLS per module, 1000M/100M Base-T Ethernet interface (unicast/multicast); IPv4, IPv6 output IP null packet filter
System	Web based management	Web based management	Web based management	Web based management	Web based management
System	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface	Chinese-English control interface
System	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W× L× H)	Dimension (W× L× H)	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm
Miscellaneous	Approx weight	Approx weight	5kg	5kg	5kg
Miscellaneous	Temperature	Temperature	0~45℃(work), -20~80℃Storage	0~45℃(work), -20~80℃Storage	0~45℃(work), -20~80℃Storage
Miscellaneous	Power	Power	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz	AC 100V-220V±10%, 50/60Hz

FMUSER HEVC/H.265 encoder's advantages

1. Providing smooth TS for modulators

FMUSER HEVC/H.265 encoder adopts Fujitsu chip which offers stable bitrate with lower fluctuation compared with other encoding chips, so it provides smooth TS for modulators. It is widely used in variety of digital distribution systems such as CATV digital head-end, satellite and terrestrial digital TV, etc.

2. Encoding with highest compression format—B frame (IBBP)

What is B Frame?

There are 3 major picture types used in the different video algorithms, they are I, P and B.

They are different in the following characteristics:

Iframes are the least compressible but don't require other video frames to decode.

Pframes can use data from previous frames to decompress and are more compressible than Iframes.

Bframes can use both previous and forward frames for data reference to get the highest amount of data compression.

Frame Type	Byte of data/KB	Compression Ratio
I	18	7:1
P	6	20:1
B	2.5	50:1

In one word, B frame is the highest compression format which makes it possible to process HD video at low bitrate. HEVC/H.265 encoder is not able to save bandwidth unless it is with B frame. In encoder parameters, B frame is often described in GOP (Group of Pictures) structure, like "IBBP".

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)